Giuseppe Mignemi

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Research interests

Bayesian Nonparametric Measurement Models

Flexible Rater-Mediated Assessment

Bayesian Multilevel Models

Current position

December 2025 - Present
Assistant Professor of Psychometrics and Measurement Models
Department of Social Sciences
Kore University of Enna

Former positions

October 2025 - December 2025
Postdoc Research Fellow,
Department of Statistical Modelling (ICS)
Czech Academy of Sciences
Research supervisor: Patricia Martinkova

February 2024 - July 2025
Postdoc Research Fellow,
Bocconi Institute for Data Science and Analytics
Bocconi University
Research supervisors: Antonio Lijoi and Igor Prünster

Education

October 2020 - December 2023
University of Padova - Italy
PhD in Psychometrics, [PhD course in *Psychological Sciences*, XXXVI cycle]
PhD supervisors: Andrea Spoto and Antonio Calcagnì
Accepted *cum laude* [Defense: 15th March 2024].

October 2018 - April 2020
University of Padova - Italy
MSc in Social, Work and Communication Psychology
Graduation grade: 110/110 *cum laude*.

September 2015 - July 2018
University of Catania - Italy
BSc Psychological Sciences and Techniques
Graduation grade: 110/110 *cum laude*.

September 2021 - July 2022 **University of Padova (Statistics Department)** - Italy
Selected courseworks
Theory and Method of Statistical Inference, Computational Statistics, Data mining.

Honors and scholarships

2026 **Marie Skłodowska-Curie Actions - Seal of Excellence**.
Quality label awarded under Postdoctoral Fellowships for the project: *BABEL - BAyesian Biclustering models for hEterogeneous Learning*
[Supervisor: P. Martinkova, Co-supervisors: E. Erosheva, A. Lijoi].

2025 **EAM Fellowship for Young Researchers**.
EAM Mobility Grants for Young Researchers provided by the European Association of Methodology for the project: *Robust Confirmatory Factor Analysis*.

2025 **Early-career Researchers PPLZ Post-doctoral Fellowship**.
Postdoctoral Research Fellow supported by the Czech Academy of Sciences (PPLZ), conducting independent research within an international academic environment for the project: *Capturing Individual Latent Heterogeneity via Flexible Bayesian Models* [Two Years Fundings - not taken].

2025 **BNP-ISBA Junior Travel Award**.
Travel support awarded by the Section on Bayesian Nonparametrics of the International Society for Bayesian Analysis. *BNP-14 World Meeting*, UCLA, Los Angeles (USA), 23-27 June 2025.

2020-2023 **Merit-based PhD Fellowship.**
General Psychology Department - University of Padova.

2020 **Mille e una Lode Scholarship**.
Award for the top 3% best students of each university course.

Academic Visiting Periods

September 2025 **Generalized Infinite Hidden Markov Models**
Mentor: Francesco Bartolucci, Economy Department, University of Perugia.

December 2024 **Bayesian Nonparametric Rater-Mediated Assessment Models**
Mentor: Patricia Martinkova, Institute of Computer Science, Czech Academy of Science.

December 2022 - March 2023 **Statistical modelling of peer grading**
Mentors: Irini Moustaki and Yunxiao Chen, London School of Economics.

September 2022 - December 2022 **Mixture polarization in inter-rater agreement analysis: a Bayesian nonparametric index**
Mentor: Ioanna Manolopoulou, University College London.

Teaching experience

February 2026 - May 2026 — **Professor, *Quantitative Assessment Techniques* [54 hours]: graduate course in Psychology - (Kore University)**
Classical Test Theory, Factor Models, Reliability Analyses, Human-in-the-Loop approaches for Psychometric Scale Development.

February 2026 - May 2026 — **Professor, *Test Theory and Techniques* [36 hours]: undergraduate course in Psychological Science - (Kore University)**
Item Response Theory Models (1-4 PL), Rater-Mediated Measurement Models, Quantitative Techniques and Standards for Reliable, Valid and Fair Measurements.

March 2024 - June 2024 — **Teaching Assistant, *Introduction to statistical methods in psychology* [25 hours]: undergraduate course in Psychological Science - PS2648 (University of Padova)**
Linear Models and Latent Variables Models for Behavioral Sciences.

March 2023 - June 2023 — **Teaching Assistant, *Introduction to statistical methods in psychology* [20 hours]: undergraduate course in Psychological Science - PS2648 (University of Padova)**
Linear Models and Latent Variables Models for Behavioral Sciences.

September 2021 - February 2022 — **Teaching Assistant, *Basic research methods* [8 hours]: undergraduate course in Psychological Science - PS2192 (University of Padova)**
Data cleaning, Descriptive Statistics and Coding with R.

September 2021 - February 2022 — **Teaching Assistant, *Psychometrics* [10 hours]: undergraduate course in Psychological Science - PS2192 (University of Padova)**
Statistical Coding and Data Analysis for Psychological Sciences.

September 2021 - December 2021 — **Teaching Assistant, *Psychological testing* [20 hours]: undergraduate course in Psychological sciences of development, personality and interpersonal relationships - PS2295 (University of Padova)**
Basic Inferential Statistics and Linear Algebra for Psychometric Models.

Communication at conferences and meetings (selected)

26 August 2026 — **Mignemi, G.**, Tancini, D., Bartolucci, F.
Extended Infinite Hidden Markov Models.
8th MBC2 Workshop, Catania, Italy

15 April 2026 — **Mignemi, G.**, Martinkova, P., Paganin, S.
Dynamic Extended Generalized Many-Facet Rasch Model
15th International Conference on Multilevel Analysis, Utrecht, the Netherlands

5 February 2026 — **Mignemi, G.**, Martinkova, P.
Discrete Nonparametric Priors in Confirmatory Factor Analysis: Within Group Invariance Analysis
Psychoco Workshop, Padova, Italy

22 September 2025 — **Mignemi, G.**, Manolopoulou, I.
Bayesian Nonparametric Linear and Generalized Linear Models for Inter-Rater Reliability Analysis [invited talk]
Department of Economy, University of Perugia, Italy

09 September 2025 — **Mignemi, G.**, Manolopoulou, I.
Flexible Models for Multiple Raters Data via Nonparametric Priors [invited talk]
15th Scientific Meeting Classification and Data Analysis Group, Napoli, Italy

3 September 2025 — **Mignemi, G.**, Panzeri, A., Bruno, G., Spoto, A.
Infinite Dimensional Psychology: Bayesian Nonparametrics in Psychology
European Mathematical Psychology Group meeting, Padova, Italy

26 June 2025 — **Mignemi, G.**, Lijoi, A., Pruenster, I.
Nonparametric Estimation of Species Richness via Penalized Gibbs-type Priors
14th International Conference on Bayesian Nonparametrics, Los Angeles, USA

27 May 2025 — **Mignemi, G.**, Lijoi, A., Pruenster, I.
Estimation of Species Richness via Penalized Gibbs-type Priors [Poster]
Bayesian Inference in Stochastic Processes workshop, Milan, Italy

20 March 2025 — **Mignemi, G.**, Manolopoulou, I.
Bayesian Nonparametric Generalized Rater Models [invited talk]
Centre for Educational Measurement - University of Oslo - Oslo, Norway.

5 December 2024 — **Mignemi, G.**, Manolopoulou, I.
Bayesian Nonparametric Models for Multiple Raters: a General Statistical Framework [invited talk]
Computational Psychometrics Group - Czech Academy of Sciences - Prague, Czech Republic.

16 July 2024 — **Mignemi, G.**, Manolopoulou, I.
Bayesian nonparametric HETOP models for rating data
89th annual meeting of the Psychometric Society. Prague, Czech Republic.

20 June 2024 — **Mignemi, G.**, Chen, Y., Moustaki, I.
Statistical Analysis of Peer Grading: A Latent Variable Approach
52st scientific meeting of Italian Statistical Society - Bari, Italy.

23 November 2023 — **Mignemi, G.**, Chen, Y., Moustaki, I.
Network of Peer Gradings: A Latent Variable Approach [invited talk]
Social Statistics Lab - London School of Economics - London, UK

11 September 2023 — **Mignemi, G.**, Manolopoulou, I., Calcagnì, A.
Groups heterogeneity in classification tasks: a Bayesian nonparametric approach
Cladag 2023 - Salerno, Italy.

10 July 2023 **Mignemi, G.**, Manolopoulou, I.
Inter-rater reliability in heterogeneous samples: a Bayesian semi-parametric approach [Poster]
Greek Stochastics ν′, Contemporary Bayesian Inference, Naxos, Greece, 7-10 July 2023.

27 January 2023 **Mignemi, G.**, Manolopoulou, I.
Inter-rater Agreement and cluster polarization: a Bayesian semi-parametric approach [invited talk]
Social Statistics Lab - London School of Economics - London, UK

15 July 2022 **Mignemi, G.**, Calcagnì, A., Spoto, A.
A latent thresholds model in rating tasks and inter-rater agreement
87th annual meeting of the Psychometric Society. Bologna, Italy.

24 June 2022 **Mignemi, G.**, Manolopoulou, I., Spoto, A., Calcagnì, A.
Latent thresholds model in classification tasks
51st scientific meeting of Italian Statistical Society - Caserta, Italy.

21 November 2021 **Mignemi, G.**
The Power of Statistical Power: the Sample Size in Social Psychology [invited talk]
SpecoLab, DPSS, University of Padova, Italy

Publications

In preparation **Mignemi, G.**, Martinkova, P., Exploring Multi-way Measurement Invariance via Discrete Nonparametric Priors in the Hybrid age of AI.

Mignemi, G., Tancini, D., Bartolucci, F. Generalized Infinite Markov Models for Heterogeneous Latent Transition Analysis.

Mignemi, G., Chen, Y., Calcagni, A., Moustaki, I. Robust CFA in the Presence of Unknown Correlated Measurement Errors.

Mignemi, G., Infinite Dimensional Measurement: Introduction to Discrete Nonparametric Priors for Psychometric Analysis.

Lijoi, A., **Mignemi, G.**, Prunster, I. Bayesian estimation of species richness with penalized Gibbs-type priors. [5]

Mignemi, G., Martinkova, P., Paganin, S. Dynamic Extended Generalized Many-Facet Rasch model. [6]

Under Review **Mignemi, G.**, Bottaro, R., Faraci, P. Do You Shrink or Penalize? Comparison between Penalized- and Shrinkage-Prior- Structural Equation Models.

Published **Mignemi, G.**, Manolopoulou, I. Bayesian Nonparametric Models for Multiple Raters: a General Statistical Framework. ***Psychometrika***, Published online, 1-36. [8]

Mignemi, G., Chen, Y., Moustaki, I. (2025) Unfolding the Network of Peer Grades: A Latent Variable Approach. ***Psychometrika***, 90(3), 1153-1174. [9]

Mignemi, G., Chen, Y., Moustaki, I. (2025). Statistical Analysis of Peer Grading: A Latent Variable Approach. In: *Proceedings from 52st scientific meeting of Italian Statistical Society (SIS)*, 22-24 June 2024, Bari, Italy. [10]

Mignemi, G., Calcagnì, A., Spoto, A., Manolopoulou, I. (2024). Mixture polarization in inter-rater agreement analysis: a Bayesian nonparametric index. ***Statistical Methods and Applications***, 33, 325–355.[11]

Mignemi, G., Manolopoulou, I., Calcagnì, A. Groups heterogeneity in classification tasks: a Bayesian nonparametric approach. In: *Proceedings from 14th Scientifc Meeting of the Classification and Data Analysis Group (Cladag)*, Salerno, September 11-13, 2023. ISBN: 9788891935632. [12]

Mignemi, G., Manolopoulou, I., Spoto, A., Calcagnì, A. Latent thresholds model in classification tasks. In: *Proceedings from 51st scientific meeting of Italian Statistical Society (SIS)*, 22-24 June 2022, Caserta, Italy. ISBN 9788891932310. [13]

Psychometric Analysis in Social Sciences

Vicovaro, M., **Mignemi, G.**, Nucci, M., Bolognani, L., Iannattone, S., Bruno, G., and Spoto, A. (2025). A new Bayesian method for the estimation of emergency nurses' thresholds and agreement in the context of telephone triage. *Frontiers in Psychology*, 16, 1477844. [14]

Riccardi, A., Barbadoro, F., Margoni, M., **Mignemi, G.**, Pucci, V., Mondini, S., ... and Gallo, P. (2024). Neuropsychological Awareness in Pediatrics (NAP): proposal of a new cognitive self-report questionnaire for pediatric Multiple Sclerosis patients. *Multiple Sclerosis and Related Disorders*, 106251. [15]

Iannattone, S., **Mignemi, G.**, Pivetta, E., Gatta, M., Sica, C., Cardi, V., Canale, N., Spoto, A., Bottesi, G. (2024). Are anxiety, depression, and stress distinguishable in Italian adolescents? an examination through the Depression Anxiety Stress Scales-21. *PloS one*, 19(2), e0299229. [16]

Panzeri, A., **Mignemi, G.**, Bruno, G., Granziol, U., Scalavicci, C., Bertamini, · Marco, Bennett, K. M., Spoto, A., Vidotto, G. (2023). The role of the COVID-19 impersonal threat strengthening the associations of right-wing attitudes, nationalism and anti-immigrant sentiments. *Current Psychology*, 1, 3. [17]

Mignemi, G., Panzeri, A., Granziol, U. et al. (2022). The mediating role of scientifical-medical satisfaction between COVID-19 conspiracy beliefs and vaccine confidence: a two-waves structural equation model. *Journal of Behavioral Medicine*,46, 201–211. [18]

Panzeri, A., Bettinardi, O., Bottesi, G., Bertolotti, G., Brambatti, G., Monfredo, M., **Mignemi, G.**, Bruno, G., Vidotto, G., Spoto, A., Frattola, P., Chiesa, S. (2022) Assessment of perceived support in the context of emergency: Development and validation of the psycho-social support scale. *Current Psychology*. [19]

Shevlin, M., Butter,..., **Mignemi, G.**, Spoto, V., Vidotto, G., Bentall, R. P. (2022). Measurement invariance of the Patient Health Questionnaire (PHQ-9) and Generalized Anxiety Disorder scale (GAD-7) across four European countries during the COVID-19 pandemic. *BMC psychiatry*, 22(1), 154. [20]

Maeran, R.,**Mignemi, G.** (2022). *Psicologia del turismo, nuove forme interattive di consumo*. Bologna: Patron Editore. [21]

Maeran, R., **Mignemi, G.** (2021). The role of consumer-brand engagement in tourism consumption: from user-generated content to my generated content. *Testing, Psychometrics, Methodology in Applied Psychology*, Vol. 28, pp. 505-520. [22]

Open packages and software

RatersBNP R package for the flexible analysis of rater-mediated assessments. It implements the method detailed in [8].
Github repository

GibbsSpecies R package for the species richness estimation via mixtures of symmetric Dirichlet distributions. It allows for the direct implementation of the method proposed in [5].
Github repository

Service to profession

Referee: *Psychometrika*

Journal of Educational and Behavioral Statistics

British Journal of Mathematical and Statistical Psychology

Psychological Methods

Behavior Research Methods

Frontiers in Psychology: Quantitative Psychology and Measurement

Editorial Board: *Turismo e Psicologia* [Padova University Press; 2020-2025] (Quantitative Methods Consultant)

Research network

Membership: *COMputational PSychometrics Group* (ICS-CAS; P. Martinkova)

BayesLab (Bocconi University; I. Pruenster)

Psychometric Lab (LSE; I. Moustaki)

Quantitative Psychology Lab (University of Padova; A. Spoto)

Memberships in scientific societies

2026 – Present National Council on Measurement in Education

2025 – Present Association for Psychological Sciences

2025 – Present European Association of Methodology

2024 – Present International Society of Bayesian Analysis

2022 – Present Italian Statistical Society

2022 – Present Psychometric Society

2022 – Present Associazione Italiana di Psicologia

Technical skills

Programming languages
Familiar with: C, C++, Python.

Software
LaTeX, R, Mathematica, STAN, Nimble, MATLAB.

Languages
English (fluent), French (basic), Spanish (basic).